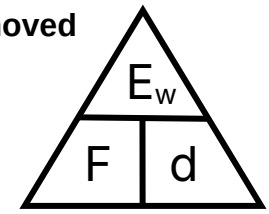
Energy Work done = Force applied x distance moved



- Force (F) is measured in Newtons (N)
- Distance (d) is measured in metres (m)
- Energy Work (E_w) done is measured in Joules (J) or Nm Newton metre (Nm)

Example

A lift raises a mass of 1000kg to the next floor in the Union Square shopping centre, which is 20m up. Calculate the minimum amount of work that must be done by the winch.

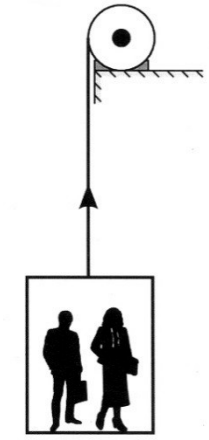
Remember – Gravitational field strength is always 9.8 N/kg. The gravitational field strength is provided in the data booklet as 9.8 ms^{-2} (don't worry about the units as it is the same).

Convert the weight of lift into a force = mg (mass x gravitational field strength**)**

What are my knowns?
m = 1000kg
d = 20m
g = 9.8 N/kg

F = mg
= 1000 x 9.8
= 9800 N

E_w = Force x distance
= 9800 x 20
= **196,000 J or 196KJ**



Rounding and significant figures

When answering calculations on this course, make sure you are using significant figures within your final answer. In a number, all figures are significant, except zeros at the front (to show decimal points) or at the end. For example, 1234 has four significant figures. 0.012 has only two significant figures, as does 1200. You should also take care to round your numerical answers to an appropriate number of significant figures.

$3.6 \times 10^{3} = 3600.00$ $3.6 \times 10^{-3} = 0.0036$

Questions

(Answer to 2 significant figures)

1. Calculate the amount of work done when lifting the following masses to a height of 2m:

   a) 0.5kg

   E_w = F x d F = ma = 0.5 × 9.8 = 4.9N

   = 4.9 × 2

   = **9.8 J** to two significant figures

   b) 2. 1kg

   E_w = F x d F = ma = 2.1 × 9.8 = 20.58N

   = 20.58 × 2

   = 41.16 J = **41 J** to two significant figures

   c) 3. 2kg

   E_w = F x d F = ma = 3.2 × 9.8 = 31.36N

   = 31.36 × 2

   = 62.72 J = **63 J** to two significant figures

   d) 3.5kg

   E_w = F x d F = ma = 3.5 × 9.8 = 34.3N

   = 34.3 × 2

   = 68.6 J = **69 J** to two significant figures

2. Calculate the amount of work done when a force of 150N is used to pull a 50kg bag of sand 20 metres.

   E_w = F x d

   = 150 × 20

   = **3,000 J**

3. During the loading process, a forklift truck lifts a pallet of bricks with a mass of 740kg to a height of 2m.

    a. Calculate the minimum amount of work the truck must do during the lift.

    E_w = F x d F = ma = 740 × 9.8 = 7252N

    = 7252 × 2

    = 14,504 J = **15,000 J** = **15kJ** to two significant figures

    b. Can you also suggest why the actual work done will be greater than this?

    Weight of lifting apparatus (forks etc), air resistance, overcome friction in machine.

4. A shopping trolley is pushed for a distance of 25m. The average force applied is 400N. Calculate the work done.

    E_w = F x d

    = 400 × 25

    = **10,000 J = 10kJ**

5. Gardener Jones mows his lawn using a lawnmower with an average force of 230N over a distance of 42m. Calculate the work done.

    E_w = F x d

    = 230 × 42

    = **9,660 J = 9,700 J = 9.7kJ**

6. A car is brought to rest over a distance of 24m. The frictional force is 980N. Calculate the work done in bringing the car to rest.

    E_w = F x d

    = 980 × 24

    = **23,520 J = 24,000 J = 24kJ**

7. A mother pushes her baby in a pram for a distance of 768m. The total work done is 210kJ. Calculate the average force applied.

$E_w = F \times d$

$F = E_w / d$

$= 210{,}000 / 768$

$=$ **273.4375N = 270N** to two significant figures

8. A car tows a caravan for a distance of 12km. The total work done is 30MJ. Calculate the average force applied.

$E_w = F \times d$

$F = E_w / d$

$= 30{,}000{,}000 / 12{,}000$

$=$ **2,500N = 2.5kN** to two significant figures

9. A tug boat is towing a small oil rig. The boat produces 205MJ of work by exerting an average force of 15kN on the oil rig. Calculate the distance the oil rig is towed.

$E_w = F \times d$

$d = E_w / F$

$= 205{,}000{,}000 / 15{,}000$

$=$ **13,666.6m = 14,000m = 14km** to two significant figures

10. A person of mass 50kg runs up a flight of stairs. The vertical height of the stairs is 8m. Calculate:

    a) The weight of the person

    Weight = ma = $50 \times 9.8 = 490$N

    b) Work done

    $E_w = F \times d$

    $= 490 \times 8$

    $=$ **3,920 J = 3,900 J = 3.9kJ** to two significant figures